FOR A COCARTESIAN SQUARE OF LOCAL COMPLETE INTERSECTION MAPS THE KERNEL OF THE COTANGENT COMPARISON IS FINITE PROJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The cotangent complex of a local complete intersection map being perfect of tor-amplitude in [-1, 0]. invoked at proof lines 22896-22897. Is the structural input identifying the two complexes as perfect with bounded amplitude.
- Permanence of perfectness and of tor-amplitude under base change and the behaviour of these properties under distinguished triangles. invoked at proof lines 22898-22908. Is what makes the cone C perfect with the stated amplitude.
- The base change computation for the cotangent complex, identifying the cohomology of the cone. invoked at proof lines 22908-22910. Locates the single nonzero cohomology module of C.

Source reference. Original lemma 00NX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 875–875. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 8338: sources/stacks-more-algebra/pinned-original-source.tex, source lines 8338–8338. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 22771: sources/stacks-more-algebra/pinned-original-source.tex, source lines 22771–22771. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 16972: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16972–16972. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 22856: sources/stacks-more-algebra/pinned-original-source.tex, source lines 22856–22856. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 16003: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16003–16003. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 19017: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19017–19017. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 17148: sources/stacks-more-algebra/pinned-original-source.tex, source lines 17148–17148. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 22754: sources/stacks-more-algebra/pinned-original-source.tex, source lines 22754–22754. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 16997: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16997–16997. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 18949: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18949–18949. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Consider a cocartesian diagram of rings*

$$\begin{array}{ccc} B & \longrightarrow & B' \\ \uparrow & & \uparrow \\ A & \longrightarrow & A' \end{array}$$

If $A \to B$ and $A' \to B'$ are local complete intersections as in Definition 8338, then the kernel of $H^{-1}(NL_{B/A} \otimes_B B') \to H^{-1}(NL_{B'/A'})$ is a finite projective B'-module.

Proof. By Lemma 22856 the complexes $NL_{B/A}$ and $NL_{B'/A'}$ are perfect of tor-amplitude in $[-1, 0]$. Combining Lemmas 22754, 19017, and 17148 we have $NL_{B/A} \otimes_B B' =$

$NL_{B/A} \otimes^{\mathbf{L}}_B B'$ and this complex is also perfect of tor-amplitude in $[-1, 0]$. Choose a distinguished triangle

$$C \to NL_{B/A} \otimes_B B' \to NL_{B'/A'} \to C[1]$$

in $D(B')$. By Lemmas 18949 and 16972 we conclude that C is perfect with tor-amplitude in $[-1, 1]$. By Lemma 22771 the complex C has only one nonzero cohomology module, namely the module of the lemma sitting in degree -1. This module is of finite presentation (Lemma 16003) and flat (Lemma 16997). Hence it is finite projective by Algebra, Lemma 00NX. □